

Shuo Guo

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Education

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) Shenzhen, China
B.Sc. in Statistics, School of Data Science Sept 2025 – May 2029 (expected)

Experience

Research Assistant May 2026 – Present
Working with PhD student in the research group of Prof. Ruoyu Sun, CUHK-Shenzhen. Currently involved in a project on optimization for deep learning, with a focus on optimizer benchmarking on NPUs.

Research Interests

- Optimization theory and optimization for machine learning
- Machine learning from a mathematical perspective

Selected Independent Study / Reading

- **Yurii Nesterov**, *Lectures on Convex Optimization*
Studied topics including convexity, gradient descent, and strongly convex complexity bounds. I'm especially interested in analyzing the convergence rate of GM, NM, accelerated GM.
- **Avrim Blum, John Hopcroft, and Ravindran Kannan**, *Foundations of Data Science*
Read selected chapters related to SVD, random graphs and VC dimension
- **Ruoyu Sun**, *Lectures on Deep Learning Optimization*
Studied topics including FCN, backpropagation, optimization algorithms in DL, initialization, normalization layers and loss landscape
- **B. Oksendal**, *Stochastic Differential Equations: An Introduction with Applications*
Currently reading. Studied topics including basic measure theory, Brownian motion and Ito's formula.

Relevant Coursework

- Probability and Statistics
- Linear Algebra (Honor)
- Calculus (Honor)
- Computer Science
- Stochastic Analysis

Academic Activities

- Participated in seminars, group meetings, and independent study sessions related to optimization algorithms, machine learning, and statistics.

Technical Skills

- Python
- Basic programming for mathematical and statistical computing
- Codex, GitHub Copilot, and other AI-assisted coding tools